SR UNIVERSITY

School of Computer Science and Artificial Intelligence

Lab Assignment # 14

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Task Description 1 – (Landing Page Design)

- **Create a simple landing page for a product or service. Include a header, hero section, and footer.**

Requirements:

- **Use HTML and CSS to structure the page.**
- **Use AI to suggest layout alignment and typography.**
- **Ensure the page is responsive across devices.**

Prompt: Create a simple landing page for an online product service. The page should include a header with a logo and navigation menu, a main section describing the service with headings, paragraphs, and a call-to-action button, and a footer with contact information and copyright details. Use appropriate HTML5 semantic tags such as <header>, <nav>, <main>, <section>, and <footer> to structure the webpage

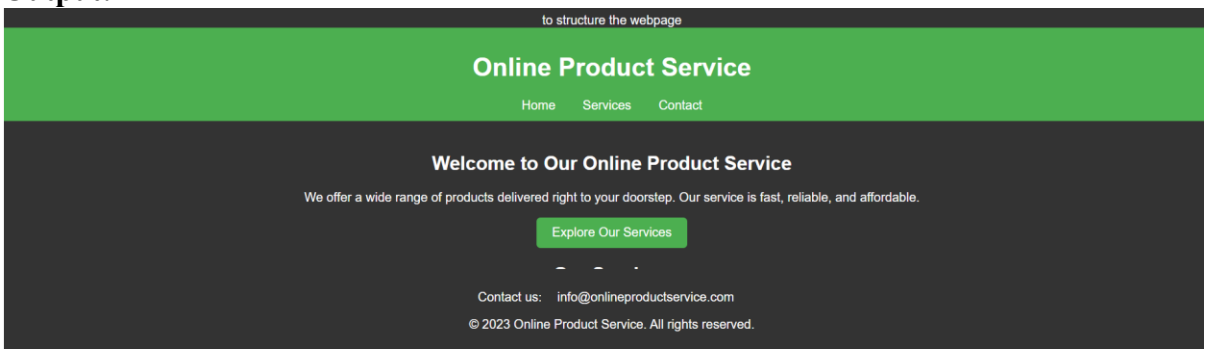
Code:-

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial
    <title>Online Product Service</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            margin: 0;
            padding: 0;
        }
        header {
            background-color: #4CAF50;
            color: white;
            padding: 10px 0;
            text-align: center;
        }
        nav {
            margin-top: 10px;
        }
        nav a {
            color: white;
            margin: 0 15px;
            text-decoration: none;
        }
        main {
            padding: 20px;
        }
        .cta-button {
            display: inline-block;
            padding: 10px 20px;
            background-color: #4CAF50;
            color: white;
            text-decoration: none;
            border-radius: 5px;
        }
        footer {
```

```
        }
        main {
            padding: 20px;
        }
        .cta-button {
            display: inline-block;
            padding: 10px 20px;
            background-color: #4CAF50;
            color: white;
            text-decoration: none;
            border-radius: 5px;
        }
        footer {
            background-color: #333;
            color: white;
            text-align: center;
            padding: 10px 0;
            position: fixed;
            width: 100%;
            bottom: 0;
        }
    </style>
</head>
<body>
    <header>
        <h1>Online Product Service</h1>
        <nav>
            <a href="#home">Home</a>
            <a href="#services">Services</a>
            <a href="#contact">Contact</a>
        </nav>
    </header>

    <main>
        <section id="home">
            <h2>Welcome to Our Online Product Service</h2>
            <p>We offer a wide range of products delivered right to your doorstep. Our service
            <a href="#services" class="cta-button">Explore Our Services</a>
        </section>
```

```html
    </style>
</head>
<body>
    <header>
        <h1>Online Product Service</h1>
        <nav>
            <a href="#home">Home</a>
            <a href="#services">Services</a>
            <a href="#contact">Contact</a>
        </nav>
    </header>

    <main>
        <section id="home">
            <h2>Welcome to Our Online Product Service</h2>
            <p>We offer a wide range of products delivered right to your doorstep. Our service is f
            <a href="#services" class="cta-button">Explore Our Services</a>
        </section>

        <section id="services">
            <h2>Our Services</h2>
            <p>We provide a variety of products including electronics, clothing, home goods, and mo
        </section>
    </main>

    <footer>
        <p>Contact us:
            <a href="mailto:info@onlineproductservice.com">info@onlineproductservice.com</a>
        </p>
        <p>&copy; 2023 Online Product Service. All rights reserved.</p>
    </footer>
</body>
</html>
```

Output:-



Justification:-
Showcases a product section with clear headings and bullet points to highlight key features and benefits for users. Focuses on readable layout and persuasive copy to improve user engagement and conversion.

Task Description 2 – (Theme Toggle Feature)
• Provide a button to switch between light and dark themes..
Requirements:
• Use JavaScript to toggle CSS classes.
• Use AI assistance to manage color contrast and accessibility.
• Ensure theme preference applies instantly on click.

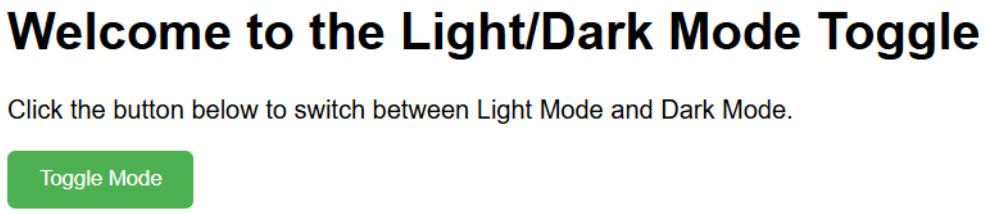
Prompt: create a webpage that displays a dismissible notification banner at the top using HTML, CSS, and JavaScript. The banner should include a message and a close button. Use JavaScript to hide the banner when the user clicks the close button and apply CSS for smooth transitions and a clean UI design.

Code:-

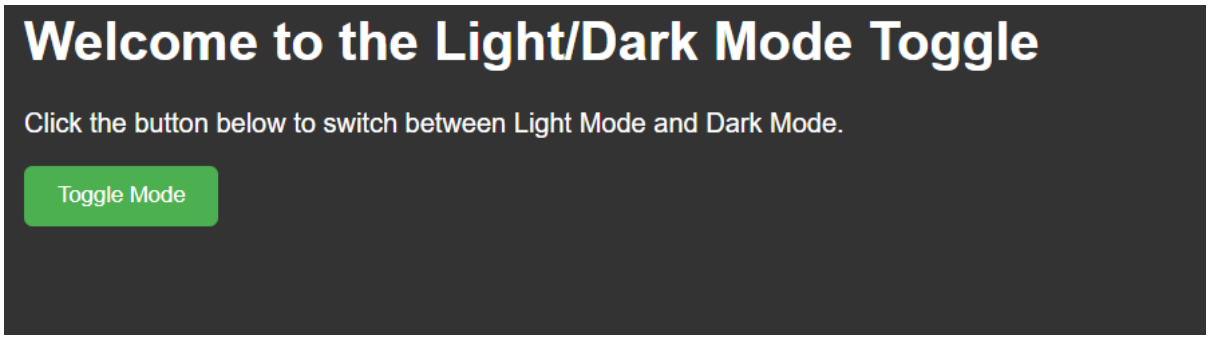
```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Light/Dark Mode Toggle</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            margin: 0;
            padding: 20px;
            transition: background-color 0.3s, color 0.3s;
        }
        .light-mode {
            background-color: #ffffff;
            color: #000000;
        }
        .dark-mode {
            background-color: #333333;
            color: #ffffff;
        }
        .toggle-button {
            padding: 10px 20px;
            background-color: #4CAF50;
            color: white;
            border: none;
            border-radius: 5px;
            cursor: pointer;
        }
    </style>
</head>
<body class="light-mode">
    <h1>Welcome to the Light/Dark Mode Toggle</h1>
    <p>Click the button below to switch between Light Mode and Dark Mode.</p>
    <button class="toggle-button" onclick="toggleMode()">Toggle Mode</button>

    <script>
        function toggleMode() {
```

```html
    <script>
        function toggleMode() {
            const body = document.body;
            body.classList.toggle('light-mode');
            body.classList.toggle('dark-mode');
        }
    </script>
</body>
</html>
```

Output:-
In white Theme



In Black Theme



Justification:-
Implements a theme toggle button that instantly switches between light and dark modes using class-based styling. Improves accessibility and user comfort by adjusting colors dynamically without reloading the page.

Task Description 3 – (User Feedback Form)
• Create a feedback form with fields: Username, Rating, Comments.
Requirements:
• Use JavaScript to validate required fields.
• Display success or error messages dynamically.
• Use AI to improve form usability and layout.

Prompt: Design a customer feedback form for a website that includes username, rating using star icons, and a comment box. Use JavaScript validation to ensure the username and rating are required fields. Style the form using CSS for better usability and readability.

Code:-

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```html
!DOCTYPE html>
html lang="en">
head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-sca
    <title>Customer Feedback Form</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            background-color: #f4f4f4;
            display: flex;
            justify-content: center;
            align-items: center;
            height: 100vh;
        }
        .feedback-form {
            background-color: #fff;
            padding: 20px;
            border-radius: 5px;
            box-shadow: 0 0 10px rgba(0, 0, 0, 0.1);
            width: 300px;
        }
        .feedback-form h2 {
            margin-bottom: 15px;
        }
        .form-group {
            margin-bottom: 15px;
        }
        .form-group label {
            display: block;
            margin-bottom: 5px;
        }
        .form-group input[type="text"],
        .form-group textarea {
            width: 100%;
            padding: 8px;
```

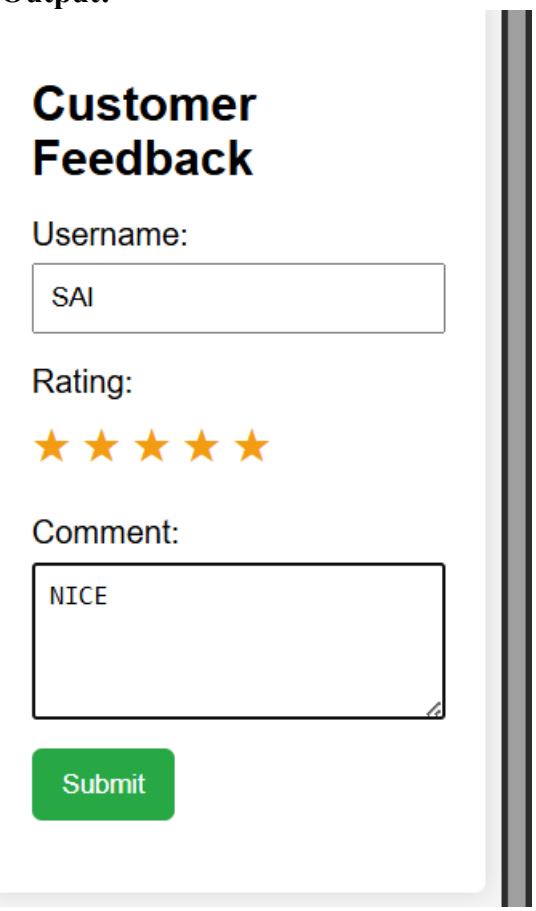
```
        padding: 8px;
        box-sizing: border-box;
    }
    .star-rating {
        display: flex;
        gap: 5px;
    }
    .star-rating input[type="radio"] {
        display: none;
    }
    .star-rating label {
        font-size: 24px;
        color: #ccc;
        cursor: pointer;
    }
    .star-rating input[type="radio"]:checked ~ label,
    .star-rating label:hover,
    .star-rating label:hover ~ label {
        color: #f39c12;
    }
    .submit-btn {
        background-color: #28a745;
        color: #fff;
        border: none;
        padding: 10px 15px;
        cursor: pointer;
        border-radius: 5px;
    }
</style>
ad>
y>
<div class="feedback-form">
    <h2>Customer Feedback</h2>
    <form id="feedbackForm" onsubmit="return validateForm()">
        <div class="form-group">
```

```html
<div class="feedback-form">
    <form id="feedbackForm" onsubmit="return validateForm()">
        <div class="form-group">
            <label for="username">Username:</label>
            <input type="text" id="username" name="username" required>
        </div>
        <div class="form-group">
            <label>Rating:</label>
            <div class="star-rating">
                <input type="radio" id="star5" name="rating" value="5"><label for="star5">
                <input type="radio" id="star4" name="rating" value="4"><label for="star4">
                <input type="radio" id="star3" name="rating" value="3"><label for="star3">
                <input type="radio" id="star2" name="rating" value="2"><label for="star2">
                <input type="radio" id="star1" name="rating" value="1"><label for="star1">
            </div>
        </div>
        <div class="form-group">
            <label for="comment">Comment:</label>
            <textarea id="comment" name="comment" rows="4"></textarea>
        </div>
        <button type="submit" class="submit-btn">Submit</button>
    </form>
</div>
<script>
    function validateForm() {
        const username = document.getElementById('username').value.trim();
        const rating = document.querySelector('input[name="rating"]:checked');

        if (username === '') {
            alert('Username is required.');
            return false;
        }
        if (!rating) {
            alert('Rating is required.');
            return false;
        }
        return true;
```

```html
            return false;
        }
        return true;
    }
</script>
</body>
</html>
```

Output:-



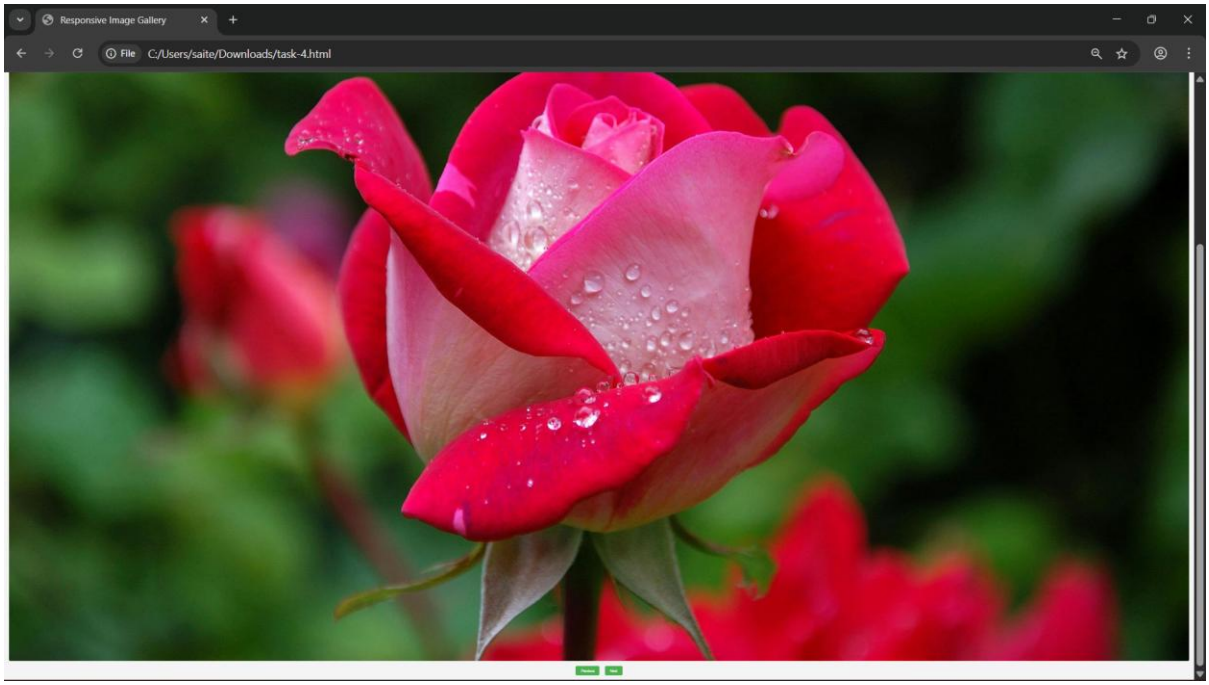
Justifications:-
Builds a feedback form capturing username, rating, and comments with JavaScript validation for required fields. Enhances usability by preventing incomplete submissions and guiding users to correct input errors.

Task Description - 4 (Image Gallery with JavaScript)
• Task: Display a small image gallery on a web page.
Requirements:
• Use JavaScript to dynamically load images.
• Add next/previous navigation buttons.
• Use AI to optimize layout responsiveness.

Prompt: Create a responsive image gallery webpage using HTML, CSS, and JavaScript. The gallery should display multiple images in a grid layout. Add Next and Previous buttons to navigate through the images dynamically using JavaScript.

// generate 6 random images to put in the gallery

Output:-



Justification:-

Creates a responsive image gallery that dynamically loads images and supports next/previous navigation buttons. Provides a clean, interactive browsing experience optimized for different screen sizes

Code:

-

Task Description - 5: (Notification Banner) • Display a notification banner at the top of the page.
Requirements:
• Use JavaScript to show and hide the banner.
• Style the banner using CSS.
• Use AI to ensure smooth transitions and clean UI behavior.

Prompt: create a webpage that displays a dismissible notification banner at the top using HTML, CSS, and JavaScript. The banner should include a message and a close button. Use JavaScript to hide the banner when the user clicks the close button and apply CSS for smooth transitions and a clean UI design.

Code:

```html
// generate 6 random images to put in the gallery
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Responsive Image Gallery</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            margin: 0;
            padding: 20px;
            background-color: #f4f4f4;
        }
        .gallery {
            display: grid;
            grid-template-columns: repeat(auto-fit, minmax(200px, 1fr));
            gap: 10px;
        }
        .gallery img {
            width: 100%;
            height: auto;
            border-radius: 5px;
        }
        .navigation {
            display: flex;
            justify-content: center;
            margin-top: 20px;
        }
        .nav-button {
            padding: 10px 20px;
            background-color: #4CAF50;
            color: white;
            border: none;
            border-radius: 5px;
            cursor: pointer;
            margin: 0 10px;
        }
```

```
    <style>
        }
    </style>
</head>
<body>
    <h1>Responsive Image Gallery</h1>
    <div class="gallery" id="gallery">
        <!-- Images will be dynamically inserted here -->
    </div>
    <div class="navigation">
        <button class="nav-button
" onclick="previousImage()">Previous</button>
        <button class="nav-button" onclick="nextImage()">Next</button>
    </div>
    <script>
        const images = [
            'https://images.pexels.com/photos/56866/garden-rose-red-pink-56866.jpeg',
            'https://images.unsplash.com/photo-1566669086984-077347c1f4bb?q=80&w=679&auto=form
            'https://plus.unsplash.com/premium_photo-1674595205685-0f6dec44acd6?q=80&w=735&aut
        ];
        let currentIndex = 0;
        const gallery = document.getElementById('gallery');
        function displayImage(index) {
            gallery.innerHTML = `<img src="${images[index]}" alt="Gallery Image">`;
        }
        function nextImage() {
            currentIndex = (currentIndex + 1) % images.length;
            displayImage(currentIndex);
        }
        function previousImage() {
            currentIndex = (currentIndex - 1 + images.length) % images.length;
            displayImage(currentIndex);
        }
        // Initial display
        displayImage(currentIndex);
    </script>
```

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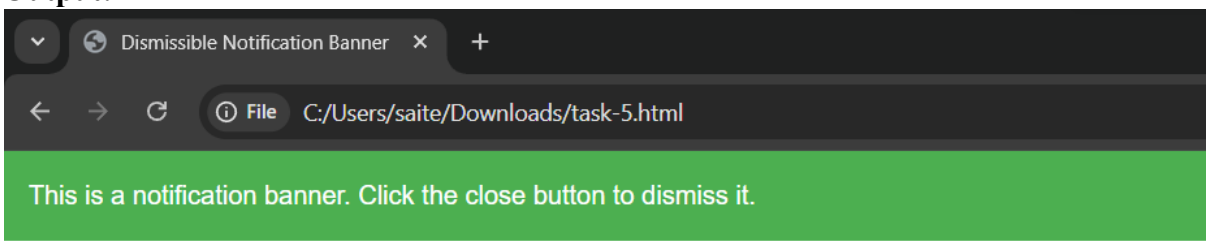
```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Dismissible Notification Banner</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            margin: 0;
            padding: 0;
        }
        .notification-banner {
            background-color: #4CAF50;
            color: white;
            padding: 15px;
            text-align: center;
            position: fixed;
            top: 0;
            width: 100%;
            display: flex;
            justify-content: space-between;
            align-items: center;
            transition: top 0.3s ease-in-out;
        }
        .close-btn {
            background-color: transparent;
            border: none;
            color: white;
            font-size: 20px;
            cursor: pointer;
        }
    </style>
</head>
<body>
    <div class="notification-banner" id="notificationBanner">
        <span>This is a notification banner. Click the close button to dismiss it.</span>
        <button class="close-btn" onclick="closeBanner()">×</button>
```

```html
        }
    </style>
</head>
<body>
    <div class="notification-banner" id="notificationBanner">
        <span>This is a notification banner. Click the close button to dismiss it.</span>
        <button class="close-btn" onclick="closeBanner()">×</button>
    </div>

    <script>
        function closeBanner() {
            const banner = document.getElementById('notificationBanner');
            banner.style.top = '-100px'; // Move the banner out of view
            setTimeout(() => {
                banner.style.display = 'none'; // Hide the banner after the transition
            }, 300); // Match the transition duration
        }
    </script>
</body>
</html>
```

Output:-



Justification:-

Displays a notification banner at the top of the page that can be shown or hidden using JavaScript. Uses smooth transitions and styled layout to deliver important messages without disrupting the overall UI.